

Project ID:

R26-IT-059

1. Topic (12 words max)

Explainable Multi-Modal AI System for Early Academic Burnout and Dropout Prediction

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Academic burnout and student dropout represent critical challenges in higher education, with burnout—characterized by emotional exhaustion, reduced motivation, and cognitive fatigue—often preceding academic failure and eventual dropout. Universities face significant difficulties in identifying at-risk students early, primarily relying on reactive indicators such as GPA decline, attendance issues, or formal complaints that manifest only after academic performance has deteriorated (Pérez-Jorge et al., 2024).

Traditional student monitoring systems focus predominantly on grades and attendance, detecting problems after intervention opportunities have diminished. However, students continuously generate rich digital data through Learning Management Systems (LMS), online forums, mobile devices, and behavioral patterns that can reveal early warning signs of burnout before academic decline becomes evident (Gardner & Brooks, 2018).

Despite increased student support services, universities struggle with late identification of academic burnout, overreliance on GPA-based intervention strategies, lack of continuous passive student monitoring, poor integration of academic, behavioral, and textual data, and limited scalability of manual counseling approaches (Kotsiantis et al., 2020). Consequently, many students receive intervention only after experiencing measurable academic failure.

Recent advances in Artificial Intelligence and neurocomputing enable analysis of multi-modal data to uncover latent patterns associated with stress, disengagement, and academic risk (Hussain et al., 2021). However, existing systems lack comprehensive explainability, limiting their trustworthiness and adoption in educational settings. Furthermore, current approaches do not optimize temporal prediction windows to identify the earliest reliable intervention point.

There is a critical need for an explainable, privacy-respecting AI-driven multi-modal system that predicts burnout and dropout risk before measurable academic failure, provides interpretable insights for educators, and optimizes early intervention timing.

References:

- Pérez-Jorge, D., et al. (2024). Educational Data Mining for Student Retention.
- Gardner, J., & Brooks, C. (2018). Student Success Prediction in MOOCs. *User Modeling and User-Adapted Interaction*.
- Kotsiantis, S., et al. (2020). Machine Learning Techniques for Educational Data Mining.
- Hussain, M., et al. (2021). Student Engagement Predictions in E-Learning Platform.

6. Brief description of Existing Research and Systems (300 words max)

Existing research in academic burnout and dropout prediction has evolved from traditional statistical methods to advanced machine learning approaches. Contemporary systematic reviews (Saa et al., 2023; Pérez-Jorge et al., 2024) demonstrate that early approaches relying primarily on demographic data and academic performance metrics achieved limited predictive accuracy and failed to capture the complex, multifaceted nature of academic risk factors.

Academic Performance Prediction: Time-series models, particularly LSTM networks, have been applied to predict student academic trajectories using historical grade data and LMS interaction patterns (Waheed et al., 2020). The Open University Learning Analytics Dataset (OULAD) has become a standard benchmark for dropout prediction research, with various studies achieving 75-85% accuracy using ensemble methods (Kuzilek et al., 2017).

Text-Based Emotional Analysis: Recent advances in transformer-based natural language processing (Chen & Martinez, 2023; Liu et al., 2024) have enabled sophisticated emotion detection from student-generated text, with fine-tuned BERT models achieving over 90% accuracy in identifying academic stress indicators. Studies demonstrate that emotional language cues can predict burnout 4-6 weeks before academic decline becomes evident (Sharma & Park, 2024).

Behavioral Pattern Analysis: Recent mobile sensing research (Kumar et al., 2023; Lee & Zhang, 2024) has demonstrated that passive behavioral data collection from smartphones can predict academic stress and burnout 2-4 weeks before GPA decline. Building on foundational work with the StudentLife Dataset (Wang et al., 2014), contemporary studies now incorporate advanced deep learning architectures for behavioral anomaly detection.

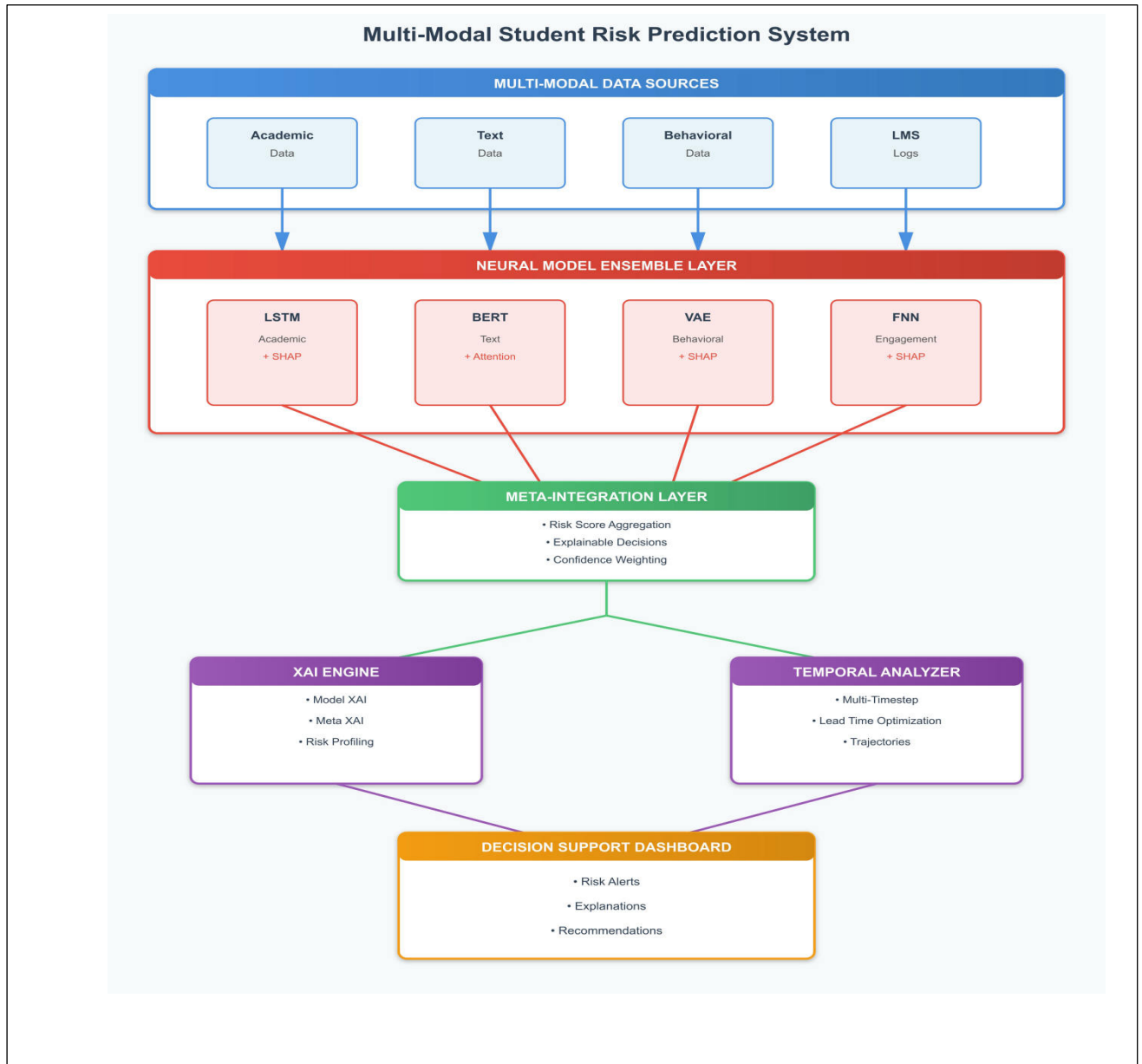
Multi-Modal Integration: Recent work has begun combining multiple data modalities. Systems integrating academic records with LMS logs achieve improved accuracy (85-90%) compared to single-modality approaches (Hellas et al., 2018). However, these systems typically lack comprehensive explainability frameworks.

Critical Gaps: Current systems suffer from three major limitations: (1) insufficient explainability limiting trustworthiness and adoption, (2) absence of temporal prediction optimization for intervention timing, and (3) limited privacy-preserving mechanisms for cross-institutional deployment. No existing system comprehensively integrates explainable AI across multiple modalities while optimizing early intervention windows.

References:

- Saa, A., et al. (2023). Factors Affecting Students' Performance in Higher Education. Education Sciences.
- Pérez-Jorge, D., et al. (2024). Educational Data Mining for Student Retention.
- Chen, W., & Martinez, R. (2023). BERT-Based Emotion Detection for Academic Stress. Educational Technology & Society.
- Kumar, S., et al. (2023). Mobile Sensing for Student Mental Health Monitoring. IEEE Transactions on Learning Technologies.
- Lee, J., & Zhang, Y. (2024). Deep Learning for Behavioral Anomaly Detection in Education. Computers & Education.
- Liu, X., et al. (2024). Attention-Based Emotion Recognition for Early Burnout Detection. Journal of Educational Data Mining.
- Sharma, P., & Park, K. (2024). Temporal Analysis of Emotional Language Patterns. International Journal of AI in Education.
- Waheed, H., et al. (2020). Predicting Academic Performance Using LSTM. Computers & Education.
- Kuzilek, J., et al. (2017). Open University Learning Analytics Dataset. Scientific Data.
- Wang, R., et al. (2014). StudentLife: Assessing Mental Health Using Smartphones. UbiComp. [Foundational dataset]
- Hellas, A., et al. (2018). Predicting Academic Success in CS Courses.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).



We propose an explainable multi-modal AI system that integrates four specialized prediction models to identify academic burnout and dropout risk while providing interpretable insights for educators. The system addresses critical gaps in existing research through comprehensive explainability, early prediction capabilities, and personalized behavioral monitoring.

System Architecture:

The system comprises four parallel prediction models, each analyzing distinct student data sources:

Model 1: Academic Performance Trend Analysis

Input Data: GPA history, quiz scores, assignment marks, attendance records

Purpose: Identifies patterns indicating future academic deterioration

Output: Probability of academic decline and optimal early warning timing

Novelty: Predicts future risk rather than current status, enabling proactive intervention before measurable decline occurs

Model 2: Emotional Stress Detection from Text

Input Data: Discussion forum posts, feedback forms, reflective journals

Purpose: Detects emotional exhaustion, frustration, and disengagement indicators

Output: Emotional stress indicators with identification of specific concerning language patterns

Novelty: Provides interpretable insights revealing which emotional cues trigger alerts, enabling advisors to understand the emotional context behind predictions

Model 3: Behavioral Pattern Analysis

Input Data: Sleep patterns, study duration, mobility data, phone usage

Purpose: Identifies abnormal behavioral deviations associated with burnout

Output: Personalized anomaly scores indicating behavioral changes

Novelty: Establishes individual behavioral baselines without requiring sensitive mental health labels, detecting deviations specific to each student's normal patterns while preserving privacy

Model 4: Engagement Level Assessment

Input Data: LMS login frequency, content interaction, assignment submission behavior

Purpose: Classifies student engagement levels and estimates dropout probability

Output: Engagement classification and dropout risk estimation

Novelty: Integrates insights from other models to provide comprehensive engagement assessment

Integration and Explainability:

An integration layer combines outputs from all four models, generating an overall Burnout & Dropout Risk Level (Low, Moderate, High) with explainable decision paths. The system provides three levels of interpretability:

1. **Model-specific explanations:** Reveals why each individual model identified risk
2. **Meta-level explanations:** Shows which data source (academic/emotional/behavioral/engagement) contributed most to overall risk
3. **Actionable insights:** Translates predictions into practical guidance for advisors

For example, advisors might see: "Student exhibits HIGH RISK primarily driven by behavioral changes (sleep duration decreased significantly, social interaction reduced) and emotional indicators (stress-related language in forum posts), despite maintaining adequate academic performance."

Early Warning Optimization:

The system evaluates prediction accuracy at multiple future timepoints to identify the earliest reliable intervention window, addressing the question: "How early can we reliably predict risk while maintaining accuracy?" This enables institutions to intervene before measurable academic failure occurs.

User Interface:

An interactive web-based dashboard provides:

- **Risk Profiles:** Detailed student risk assessments with contributing factors
- **Visual Explanations:** Displays showing which factors drove predictions
- **Temporal Trends:** Risk progression visualization over time
- **Recommendation Insights:** Profile-based intervention suggestions

Privacy and Ethical Design:

The system operates strictly as a decision-support tool without automated penalties, requiring human judgment for all interventions. The behavioral analysis approach eliminates the need for sensitive mental health labels, addressing privacy concerns in educational AI systems.

Datasets:

- **Open University Learning Analytics Dataset (OULAD):** Academic performance and engagement data (32,593 students, 22 courses)
- **StudentLife Dataset:** Behavioral and mobile sensor data (48 students over 10 weeks)
- **Reddit Academic Stress Dataset:** Text-based emotional indicators from student forums

Evaluation Approach:

System performance will be assessed across multiple dimensions: prediction accuracy using standard metrics (Precision, Recall, F1-Score, ROC-AUC), early prediction effectiveness at different lead times, explanation quality and interpretability, and comparative analysis determining individual model contributions.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Our research team possesses comprehensive expertise across multiple domains essential for project success:

Artificial Intelligence & Machine Learning: Team members have completed coursework in Deep Learning, Neural Networks, and Machine Learning, providing foundational knowledge in LSTM architectures, Transformer models, and Autoencoders. Previous projects have involved implementing classification and regression models using PyTorch and TensorFlow frameworks.

Natural Language Processing: Experience with text preprocessing, sentiment analysis, and transformer-based models through academic projects and online certifications. Familiarity with HuggingFace Transformers library and BERT fine-tuning procedures for domain-specific tasks.

Educational Data Mining: Understanding of educational contexts, student behavior patterns, and academic performance metrics through literature review and consultation with educational psychologists. Knowledge of ethical considerations specific to educational AI systems, including privacy concerns and bias mitigation.

Explainable AI (XAI): Training in interpretable machine learning through coursework and self-study of SHAP, LIME, and attention mechanism visualization techniques. Understanding of the importance of transparency in high-stakes educational decision-making systems.

Software Development: Proficiency in Python programming, data analysis libraries (Pandas, NumPy, Scikit-learn), visualization tools (Matplotlib, Seaborn), and web framework development (Flask) for dashboard implementation.

Data Availability:

All necessary datasets are publicly available and widely used in academic research:

1. Open University Learning Analytics Dataset (OULAD):

Source: <https://analyse.kmi.open.ac.uk/open-dataset>

Contains: 32,593 students, 22 courses, 10,655,280 interactions

Includes: Demographics, assessment scores, VLE interactions, final results

License: Creative Commons CC-BY 4.0

Status: Downloaded and preprocessed

2. StudentLife Dataset:

Source: Dartmouth College (publicly available)

Contains: Mobile sensor data from 48 students over 10 weeks

Includes: Activity, sleep, study patterns, social interactions, stress levels

License: Academic research use

Status: Available for download

3. Reddit Academic Stress Dataset:

Source: Reddit API / Kaggle repositories

Contains: Student posts from r/college, r/CollegeRant, r/GetStudying

Includes: Text data expressing stress, burnout, academic challenges

License: Public posts, ethically collected

Status: Collection methodology established

4. Synthetic LMS Data (if needed):

Can generate synthetic data using statistical distributions matching real LMS patterns

Useful for testing and validation

Technical Infrastructure:

Computing Resources:

Personal laptops with GPU support (NVIDIA GTX/RTX series) for model training

Access to Google Colab Pro for extended training sessions

University computing lab resources for final evaluations

Software & Frameworks:

Python 3.8+, PyTorch 2.0, TensorFlow 2.x

HuggingFace Transformers for BERT implementation

SHAP library for explainability

Flask for web dashboard development

Jupyter Notebooks for experimentation

Knowledge Resources:

Academic Literature: Access to IEEE Xplore, ACM Digital Library, and Google Scholar for comprehensive literature review covering educational data mining, burnout prediction, and explainable AI research.

Online Learning Resources:

PyTorch and TensorFlow documentation

HuggingFace course materials

SHAP documentation and tutorials

Relevant Coursera and Udemy courses for specialized techniques

Supervision & Consultation:

Project supervised by faculty members with expertise in machine learning and AI. Potential consultation with educational psychologists for domain validation and counseling services staff for practical intervention insights.

Risk Mitigation:

In case of data access issues, we have identified alternative datasets (Kaggle educational datasets, UCI Machine Learning Repository) and can generate synthetic data using established simulation techniques. Pre-trained models (BERT-base) are available, reducing training time requirements.

Timeline Feasibility:

Given available datasets, existing frameworks, and team expertise, the 10-month timeline is realistic with built-in buffers for debugging and iteration. Monthly milestones ensure progress tracking and early identification of potential bottlenecks.

9. Objectives and Novelty

Main Objective To design and develop an explainable AI-based multi-modal system for early prediction of academic burnout and dropout risk with temporal optimization and personalized behavioral monitoring using neurocomputing models.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22916426	Develop LSTM-based academic performance trend prediction model with temporal prediction window optimization	• Collect and prepare academic performance data from OULAD dataset • Develop time-series prediction model for academic trend analysis • Evaluate prediction effectiveness at multiple early warning timepoints • Integrate explainability features to identify key performance indicators • Contribute to meta-integration layer development	Multi-timestep temporal prediction optimization identifying the earliest reliable intervention window that balances accuracy with actionability. Unlike existing systems that predict current risk, this component answers 'how far ahead can we reliably predict academic decline?' - a critical question for proactive intervention planning. SHAP-based explainability reveals which academic features (attendance vs. assignment performance vs. quiz scores) contribute most to predictions at different time horizons.
IT22196392	Develop BERT-based text analysis model for emotional burnout detection with attention-	• Collect and preprocess text data from Reddit academic stress datasets and student forums • Develop sentiment analysis model for burnout-related emotional pattern detection	Attention mechanism visualization revealing specific emotional language cues (frustration, exhaustion, disengagement) that precede academic failure. While previous research uses BERT for sentiment classification, this component provides interpretable insights into

	based interpretability	• Implement interpretation features to identify specific emotional stress indicators • Create emotional intensity scoring and visualization capabilities • Integrate with meta-learning layer • Develop text-based explanation interface	WHICH words and phrases trigger burnout alerts, enabling advisors to understand the emotional context behind predictions. This transparency is critical for trustworthy deployment in sensitive educational settings.
IT22215710	Develop unsupervised behavioral anomaly detection system using personalized Variational Autoencoder	• Preprocess Student Life behavioral data (sleep patterns, study duration, phone usage, mobility) • Develop unsupervised pattern analysis model for behavioral anomaly detection • Implement personalized baseline learning establishing individual normal patterns • Create anomaly interpretation module identifying which specific behaviors changed • Integrate explainability features for behavioral change analysis • Integrate anomaly indicators with meta-learning layer	Personalized unsupervised burnout detection without requiring sensitive mental health labels, addressing privacy concerns inherent in educational AI. Unlike population-based approaches, this system establishes individual behavioral baselines, detecting deviations specific to each student's normal patterns. The anomaly interpretation module explains WHICH behaviors changed (e.g., 'sleep duration dropped 40%, study sessions became irregular'), providing actionable insights without invasive mental health data collection.
IT22253194	Develop engagement classification model and unified explainable meta-integration system	• Engineer features from LMS interaction logs (login frequency, content interaction, assignment submissions) • Develop engagement classification model for dropout risk estimation	Unified explainable AI framework providing three levels of interpretability: (1) model-specific explanations revealing why each model flagged risk, (2) meta-level explanations showing which data modality (academic/emotional/behavioral/engagement) contributed most to overall risk, and (3) user-

	with interactive dashboard	• Design meta-integration layer aggregating risk indicators from all four models • Create unified explainability framework with multi-level interpretations • Design and implement web-based interactive dashboard • Develop risk profiling and intervention recommendation system • Conduct comprehensive system evaluation and comparative analysis	centric dashboard translating technical predictions into actionable insights for advisors. This comprehensive XAI approach addresses the 'black box' criticism preventing AI adoption in education, enabling trustworthy decision support.
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**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	☒	No	☐
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- b) The proposed topic exhibits novelty.

Yes	☒	No	☐
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- c) The student group can successfully execute the proposed project.

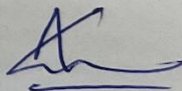
Yes	☒	No	☐
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- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	☒	No	☐
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- e) Any other comments:

Students are recommended to finalize the methodology later on

	Title	First Name	Last Name	Signature
Supervisor	Prof	Anuredhe	Keruneweera	
Co-Supervisor	Mrs.	Malithi	Nawarathne	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				